Antoine HOUNDONOUGBO

PhD in Cellular Physiology & Molecular Biology

Molecular biology · Epidemiological & genomic surveillance · Data & biostatistics · ISO 15189 quality · Training
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100,000+	500+	5	10+
PCR ASSAYS	NANOPORE GENOMES	PUBLICATIONS	STUDENTS MENTORED

01 · Profile

PhD in Cellular Physiology & Molecular Biology, with 5 years at the Hemorrhagic and Viral Fevers Laboratory (LFHV) of Cotonou (Ministry of Health, Republic of Benin), the national reference laboratory. Experience structured across five core areas: molecular diagnostics (contributed to 100,000+ PCR assays: RT-PCR, RT-qPCR, multiplex PCR for SARS-CoV-2, Lassa, Dengue, Mpox, Influenza and viral hemorrhagic fevers including Crimean-Congo, Ebola, Marburg), field genomics (500+ genomes sequenced on Nanopore and analysed by phylogenetics for national surveillance with Africa CDC and WHO), data and biostatistics (8 PhD theses and 10+ Master dissertations supported since 2019; studies of 2,043 and 1,500+ patients, R and Python, REDCap, STROBE standards), quality and compliance (ISO 15189:2022, SOPs authored and network SOPs amended with FHI 360 support), and teaching and training (10+ students supervised, Bachelor and Master teaching). Designated trainer by LFHV and FHI 360 (EpiC GHS Benin project) to deliver operational hands-on training modules in laboratory diagnostics of infectious diseases and viral hemorrhagic fevers in the two VHF laboratories of northern Benin (October 2026). Available for immediate start.

02 · Skills

Molecular biology & genomics	RT-PCR · RT-qPCR · multiplex PCR and other variants (TaqMan/SINE) · DNA/RNA extraction · Oxford Nanopore MinION sequencing · variant calling · phylogenetics (IQ-TREE, MAFFT, iTOL, Nextclade, BLAST, GISAID, GenBank) · ELISA, Luminex · cell culture · BSL-2/3 biosafety
Epidemiology, entomology & One Health	Descriptive, analytical and multi-centre studies · protocol writing and ethics submissions · KAP surveys · medical entomology: insecticide resistance (kdr, ace-1), WHO-standard bioassays, insectary rearing (Anopheles, Aedes, Culex) · rodent trapping and One Health approach
Animal health & field sampling	Microbiological surveillance project conducted with Charité - Universitätsmedizin Berlin (Germany): biological sampling in animals (blood collection, organ sampling) in rodents, bats and pangolins · trapping and handling · packaging, traceability and transport of risk samples · associated molecular (RT-PCR) and serological testing · BSL-2/BSL-3 biosafety
Data & biostatistics	R (tidyverse, dplyr, ggplot2, gtsummary, pROC) · Python (pandas, NumPy, Biopython) · REDCap, Epi Info · advanced Excel · cleaning, anonymisation, logistic and linear models, AUC ROC, composite scores, Cronbach's alpha · spatiotemporal GIS with QGIS · reporting against STROBE checklists
Quality & organisation	ISO 15189:2022 · writing and reviewing standard operating procedures (own and FHI 360 network SOPs) · IQC/EQA on Levey-Jennings charts · method validation · computerised traceability · audit preparation · multi-site planning against milestones
Teaching & communication	Lectures and practicals at Bachelor and Master level: immunodiagnosis, virology, cytology, cell regulation, medical entomology, health systems, physiology · supervision of dissertations and internships · outreach and training support · scientific writing: 5 peer-reviewed publications, 2 as first author

03 · Professional Experiences

Research Assistant - Hemorrhagic and Viral Fevers Laboratory (LFHV) of Cotonou, Ministry of Health, Republic of Benin — 08/2021 - present

- Contributed, as part of the team, to more than 100,000 PCR tests (RT-PCR, RT-qPCR, multiplex PCR) for SARS-CoV-2, Lassa, Dengue, Mpox, Influenza and hemorrhagic-fever viruses (Crimean-Congo, Ebola, Marburg) within the national epidemic-surveillance system, meeting the turnaround deadlines set by the Ministry during emergencies.
- Sequenced and analysed 500+ viral genomes (Nanopore MinION); identified circulating variants through molecular phylogenetics: data integrated into the national genomic surveillance conducted with Africa CDC and WHO.
- Coordinated a multi-centre epidemiological study on SARS-CoV-2 (2,043 patients): protocol, ethics approval, site management against milestones, REDCap database with quality controls, multivariate analyses in R, reporting; 2 first-author publications.
- Led ISO 15189:2022 compliance: authored the required SOPs, set up internal and external quality controls (Levey-Jennings), computerised traceability, audit preparation.
- Strengthened the northern-Benin hemorrhagic-fever laboratory network (FHI 360 support): reviewed and amended the SOPs drafted with the consultant hired by FHI 360, technical corrections validated; designated trainer for the technician training phase in molecular diagnostics (FHI 360 / EpiC GHS project mission, Parakou session, 19-24 October 2026).
- Supervised and trained 10+ Bachelor/Master students and laboratory staff; integrated new molecular protocols within 48 h during emergency periods.
- Supported a German research project (Charité - Universitätsmedizin Berlin) on the microbiological surveillance of rodents: contributed to organ and blood sampling in rodents, bats and pangolins; packaging, traceability and transport of risk samples; associated molecular and serological analyses.

Trainer in Biological Diagnostics of Infectious Diseases - FHI 360 · Mission at the Hemorrhagic and Viral Fevers Laboratory (LFHV), Parakou, Benin — October 2026

- Delivered hands-on training sessions in molecular biology and serology applied to infectious disease diagnostics, including viral hemorrhagic fevers, for laboratory technicians.

Field Researcher - HANTARIS “One Health” project, Institut de Recherche pour le Développement (IRD) — 09/2025 - 12/2025

- Led joint field and laboratory activities on hantavirus circulation: rodent trapping campaigns; molecular (RT-PCR) and serological analyses; dataset management and analysis; reports to the consortium.

Scientific Assistant - University Center of Anato-Cytopathology (CU-ACP), CNHU-HKM, Cotonou — 01/2022 - present

- Managed and statistically analysed clinical databases (1,500+ records) with cleaning, anonymisation and analysis protocols; retrospective study of 600 patients (clinico-endoscopic-histological correlations, logistic regression); co-author of 3 publications.
- Wrote research protocols approved by the institutional ethics committee.

Independent Consultant - Biostatistical support to student-researchers (public and private universities in Benin) — 2019 - present

- Since completing my BSc, supported researchers (PhD candidates and Master students) at URMAPha/EPAC-UAC, Faculty of Medicine of Parakou, FSS Cotonou, FAST/UAC and other institutions through the full statistical processing of their research data: KAP surveys, composite scores, Cronbach's alpha, bivariate tests, logistic modelling (adjusted ORs, AUC ROC) in R (tidyverse, gtsummary, pROC).
- Track record: 8 PhD theses and 10+ Master dissertations supported, with main contribution to the “Results” chapters: full statistical analyses, STROBE-standard figures and tables.

Intern-Researcher (medical entomology) - International Institute of Tropical Agriculture (IITA), Cotonou — 01/2021 - 08/2021

- Insecticide resistance in Anopheles gambiae, the malaria vector: 250+ high-quality DNA extractions, detection of kdr and ace-1 resistance markers by TaqMan/SINE PCR, participation in WHO-standard insecticide-susceptibility bioassays, insectary colony management.

Teaching Assistant & Lecturer - ENS-Natitingou (UNSTIM) · IRGIB Africa · École Supérieure Le Faucon · FAST, University of Abomey-Calavi 2023 - present

- Lectures and practicals at Bachelor and Master level: immunodiagnosis, virology, cytology, cell regulation, immunohematology, bioengineering, health systems, human physiology, medical entomology, animal biology; designed practicals and supervised internships.
- Academic year 2026-2027: cell regulation (Master 2 BCI, Cell Biology & Immunology, FAST, University of Abomey-Calavi); medical entomology (ABB3, École Supérieure Le Faucon).

04 · Education

PhD, Cellular Physiology & Molecular Biology - UNSTIM, Abomey · Thesis: molecular epidemiology of SARS-CoV-2 in Cotonou (NGS, phylogenetics, immune response) 2022 - 2025

MSc, Cell Biology & Immunology - FAST, University of Abomey-Calavi 2019 - 2021

BSc, Human Biology Engineering (biomedical analyses) - EPAC, University of Abomey-Calavi 2014 - 2017

05 · Certifications

- 8 certified IBM SkillsBuild badges, verifiable on Credly (as of 20/10/2026): Digital Literacy (13/08/2026) · Data Literacy (19/08/2026) · Data Classification (21/08/2026) · Data Usability for Organizations (22/08/2026) · Data Collection and Analysis (24/08/2026) · Introduction to Inferential and Descriptive Statistics (24/08/2026) · Data Preparation for Analysis (28/09/2026) · Agile Explorer (08/10/2026).
- Scientific webinars "Biosafety & Biosecurity" (attendance certificates, RCoEBB / Institut Pasteur de Dakar, 2026): ISO 35001 biorisk management system; Laboratory biosafety principles and risk management.
- In progress (2026): Python for Data Science & Statistics 101 (IBM) · Python Essentials 1 (Cisco) · ISO 15189:2022 (Alison) · Elements of AI (Univ. Helsinki) · Research ethics (TRREE) · One Health & Lab QMS (WHO).

06 · Scientific Publications

- Full publication profile and list (direct links): orcid.org/0009-0008-2051-8217
- **FIRST AUTHOR:** Houndonougbo A., Dramane G., Lagaki K., Yadouleton A. (2025). Using Next-Generation Sequencing (NGS) to Predict and Monitor the Evolution of SARS-CoV-2 Variants in Benin. *World J. Pharm. Sci. Res.* 4(1) - doi.org/10.5281/zenodo.14938943
- **FIRST AUTHOR:** Houndonougbo A., Lagaki K., Dramane G., et al. (2025). Absence of Climatic Factors Influence on the Prevalence of COVID-19 in Benin: A Spatiotemporal Analysis. *Int. J. Biosci.* 27(5), 16-23 - doi.org/10.12692/ijb/27.5.16-23
- **CO-AUTHOR:** Seidou F., Gomez D.R.H., Houndonougbo A., et al. (2025). Correlation between Clinical, Endoscopic, and Anatomopathological Aspects of Chronic Gastritis: Retrospective Study of 600 Cases at CNHU-HKM. *Open J. Pathol.* - doi.org/10.4236/ojpathology.2025.151001
- **CO-AUTHOR:** Seidou F., Gomez D.R.H., Houndonougbo A., et al. (2025). Mélanome sur nævus congénital géant chez un nouveau-né : à propos d'un cas. *J. Soc. Biol. Clin. Bénin*
- **CO-AUTHOR:** Dramane G., Lagaki A.K., Kenoukon C., Natta R., Houndonougbo A., et al. (2025). Food Preference and Inflammation: How Taste Shapes Health. *Int. J. Biosci. Bioinform.* 20(1), 1-10
- **Work under submission (4 manuscripts, titles and details available on request):** one as first author (SARS-CoV-2 seroprevalence in West Africa); three as co-author (OLGA/OLGIM staging of chronic gastritis; prostate cancer genetics and epidemiology).

07 · Languages

French (native, working language) · English (good working level) · Fon (intermediate) · Goun (native)

08 · Academic & Professional References

Pr Anges Yadouleton	DIRECT SUPERVISOR AT LFHV	Full Professor, medical entomology - ENS-Natitingou/UNSTIM; Head of the Hemorrhagic and Viral Fevers Laboratory of Cotonou (Ministry of Health)
Pr Gado Dramane	PHD SUPERVISOR	Full Professor, cell physiology - ENS-Natitingou/UNSTIM
Pr Falilatou Seidou	HEAD OF CU-ACP	Associate Professor, anatomy and cytological pathology - FSS/CU-ACP, CNHU-HKM

Reference contact details (phone, email) provided upon request.